

Fourier

Script: Chapter 6 (Fourier Series, Fourier Transform, DTFT)

Learning goals

- Compute Fourier coefficients of engine signals analytically and numerically; understand the line spectrum of the 60-2 crank signal and the effect of the missing teeth.
- Relate the decay time of a resonance to the width of its spectral peak (Lorentzian line shape, bandwidth, quality factor).
- Evaluate the DTFT by direct summation and use its properties (periodicity, symmetry).
- Use *engine orders* to distinguish speed-proportional components from fixed-frequency resonances; track orders during a speed change.

Background

A signal with period P has the Fourier series $x(\theta) = \sum_k c_k e^{j2\pi k\theta/P}$, $c_k = \frac{1}{P} \int_0^P x(\theta) e^{-j2\pi k\theta/P} d\theta$. For crank-angle signals we choose $P = 360^\circ$: then k counts oscillations per crankshaft revolution and is called the *engine order*. At speed n (rpm) order k has the frequency $f = k n/60$. Aperiodic signals have a Fourier transform $X(f) = \int x(t) e^{-j2\pi f t} dt$; a sampled signal $x[n] = x(nT_s)$ has the DTFT

$$X(e^{j\Omega}) = \sum_n x[n] e^{-j\Omega n}, \quad \Omega = 2\pi f/f_s,$$

which is 2π -periodic in Ω (f_s -periodic in f) and conjugate symmetric for real $x[n]$. For a finite record of N samples it can be evaluated at any frequency by direct summation; the FFT (Sheet 8) is only a fast way to do this on a regular grid. A sine of amplitude A whose frequency fits an integer number of periods into the record has $|X| = AN/2$, so $A = 2|X|/N$.

Exercise 6.1 Fourier series and transforms of engine signals**PAPER**

- A complete 60-tooth wheel gives a rectangular signal with period 6° and duty cycle 50%. Compute its Fourier coefficients with respect to $P = 360^\circ$. Which orders occur, what are c_0 , $|c_{60}|$, $|c_{120}|$, $|c_{180}|$? Which frequency does order 60 have at 3000 rpm?
- For the 60-2 wheel (tooth $m = 0, \dots, 57$ HIGH on $[6m, 6m + 3)^\circ$) show that

$$|c_k| = \frac{3}{360} \left| \operatorname{sinc} \frac{k}{120} \right| \left| \frac{\sin(29\pi k/30)}{\sin(\pi k/60)} \right| \quad (k \neq 60m), \quad |c_{60m}| = \frac{58}{120} \left| \operatorname{sinc} \frac{m}{2} \right|,$$

with $\operatorname{sinc} u = \sin(\pi u)/(\pi u)$. Evaluate c_0 , $|c_1|$, $|c_{59}|$, $|c_{60}|$, $|c_{61}|$ and explain the result as a “once-per-revolution modulation”. Hint: $x_{60-2} = x_{60} - x_{\text{gap}}$.

- Compute the Fourier transform of the knock model $x(t) = e^{-t/\tau} \sin(2\pi f_0 t)$, $t \geq 0$. Show that near f_0 the magnitude is a Lorentzian curve, and derive the -3 dB bandwidth Δf and the quality factor $Q = f_0/\Delta f$. Evaluate for the knock mode ($f_0 = 6.5$ kHz, $\tau = 0.8$ ms) and the valve impact (8 kHz, 0.4 ms). What follows for the bandwidth of a knock band-pass filter?
- Engine orders: list the frequencies of order 2 (combustions of the 4-cylinder engine), order 60 (crank teeth) and order 0.5 (camshaft) at 1500, 3000 and 6000 rpm, and the order at which the 6.5 kHz knock resonance appears at these speeds.

Exercise 6.2 Numerical Fourier series and DTFT**JSLINUX**

- a) The program `code/jslinux/sheet06/fourier_series.c` samples one revolution of the crank signal `es_crank()` on a grid of $M = 36000$ points (0.01°). Implement the direct summation

$$c_k \approx \frac{1}{M} \sum_{n=0}^{M-1} x[n] e^{-j2\pi kn/M}, \quad k = 0, \dots, 1200$$

(use a rotating phasor instead of calling `cos` and `sin` for every term) and compare with your results of 6.1 b).

- b) Implement the partial sum $S_K(\theta) = c_0 + \sum_{k=1}^K 2 \operatorname{Re}\{c_k e^{j2\pi k\theta/360}\}$ and evaluate it around the gap for $K = 60, 180, 600, 1200$. Report maximum, overshoot and rms error. Explain the Gibbs phenomenon: what converges and what does not?
- c) `code/jslinux/sheet06/dtfft.c`: implement the DTFT by direct summation. For a damped sinusoid ($f_0 = 6.5$ kHz, $f_s = 50$ kHz, $N = 500$) and $\tau = 0.4/0.8/1.6$ ms the program determines peak frequency, peak value and -3 dB bandwidth. Compare with 6.1 c) (note: $X_{\text{DTFT}} \approx f_s X(f)$), and compare the finite sum with the closed-form DTFT of $a^n \sin(\Omega_0 n) u[n]$.
- d) Check numerically: $|X|$ at f , $f + f_s$, $-f$ and $f_s - f$. Explain.
- e) Apply the DTFT to a real knock burst of the virtual engine (both modes, noise, combustion vibration). Identify all features of the spectrum.

Exercise 6.3 Engine orders

JSLINUX

`code/jslinux/sheet06/orders.c` records the knock-sensor signal of the virtual engine (knock probability 0.2) at constant 1500, 3000 and 6000 rpm, each over exactly 8 engine cycles (16 revolutions, $f_s = 50$ kHz), removes the mean and evaluates the amplitude spectrum $A(f) = 2|X(f)|/N$ by direct summation on 0 Hz to 1000 Hz (1 Hz steps) and 4 kHz to 12 kHz (10 Hz steps).

- a) Implement `amp_spectrum()`. Why is the record chosen as an integer number of revolutions?
- b) The combustion vibration of the model is a half-sine $0.2 \text{ V} \cdot \sin(\pi\varphi/60^\circ)$ for $0 \leq \varphi < 60^\circ$ after every TDC (period 180°). Compute its Fourier series and the amplitudes of orders 2, 4, 6, 8; enter your formula in `halfsine_order_amp()`.
- c) Run the program. Compare the low-frequency lines (frequency and amplitude) at the three speeds with b). For the high band, the program smooths $|X|^2$ over ± 150 Hz and reports the peak in three bands – tabulate frequency and order of the peaks.
- d) Explain the fine structure of the high-frequency spectrum: why does the valve resonance appear as a comb of lines with a spacing of order 2 under a Lorentzian envelope, while knock gives a smooth hump?
- e) An ECU wants to monitor a crank-synchronous vibration and the knock resonance with fixed band-pass filters. Which of the two filters must be adapted to the engine speed?

Exercise 6.4 Order tracking during a run-up

CHALLENGE

`code/jslinux/sheet06/order_track.c`: the engine accelerates from 1500 rpm to 6000 rpm within 1 s; the knock-sensor signal (no knock) is sampled at 50 kHz, crank edges are time-stamped as in Exercise 4.4.

- a) The program computes the amplitude spectrum of the time signal on 0 Hz to 500 Hz. Describe it. Why can no order amplitude be read from it?
- b) Resample the signal onto a 0.5° grid (code of 4.4) and compute the “order spectrum” $A(O) = 2|\sum_m x_a[m] e^{-j2\pi Om\Delta\theta/360^\circ}|/N_a$ for $O = 0 \dots 10$ over an integer number of revolutions. Compare the order amplitudes with Exercise 6.3.

- c) The angle grid has a “sampling rate” of $360^\circ/\Delta\theta = 720$ samples per revolution. Which orders can be represented, and which signal components alias in the angle domain at 1500 rpm? Why does this not disturb the low orders here?

Deliverables

6.1: derivations, coefficient values, Lorentzian bandwidths, order table. 6.2: completed programs, coefficient table, Gibbs table with explanation, DTFT table and the checks d)/e). 6.3: completed program, order amplitudes (theory vs. measurement) and resonance table with interpretation. 6.4 (optional): order spectrum vs. time spectrum of the run-up.